

ZK-LoRaWAN Completed Project Work & Current Checklists

Awaiting Funding Approval



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VERIFIED HARNESS

FOR: ZK-LORAWAN MAINNET & DEVNET

PHYSICAL MESH FIELD TRIAL RUNBOOKS
AND HARDENED DEPLOYMENT SAFEGUARDS

Section I: Completed Project Work

ZK-LoRaWAN Completed Work & Forensic Audit

This document inventories and explains the production-ready cryptographic, network, and smart contract modules currently built and 100% verified in the ZK-LoRaWAN repository.

1. LLD-AC Proof Compression (Component 08)

Overview

Low-Latency Decompression Arithmetic Coding (LLD-AC) is a custom serialization scheme built to compress cryptographic proofs to fit within the extremely tight physical payload limits of LoRaWAN networks (where MTU spans from 51 bytes at SF10 to 222 bytes at SF7).

Files Implemented

- `[lld_ac_encoder.py](file:///k:/Zk-LoRaWan/proof_compression/lld_ac_encoder.py)`: Implements bit-packing, coordinate normalization, and arithmetic compression.

Accomplished & Verified Features

- **Coordinate Normalization**: Maps large 256-bit G1/G2 curve points down to compact byte arrays.
- **MTU Fit Optimization**: Compresses a raw 130-byte Groth16 proof down to **108 bytes** (a 16.9% reduction).
- **Lossless Round-Trip**: Verified zero-loss reconstruction during compression/decompression cycles.

2. XOR-FEC Parity Recovery (Component 06)

Overview

A Forward Error Correction (FEC) module using exclusive-OR parity logic to automatically reconstruct dropped or corrupted proof frames over long-range, high-loss radio channels.

Files Implemented

- `[xor_fec.py](file:///k:/Zk-LoRaWan/fec/xor_fec.py)`: Codec implementing block dividing, parity generation, and frame reconstruction.

Accomplished & Verified Features

- **Data Padding & Grouping**: Pads variable-length proofs to static boundaries for matrix operations.
- **Parity Overhead Control**: Adds 2 parity blocks for every 8 data blocks (~36.7% overhead) for robust error margins.

- **Loss Recovery:** Successfully reconstructs missing data groups under single and multi-block loss simulations.

3. ZK Semantic Gating / Range Proofs (Component 02)

Overview

Gating protocol enforcing that device telemetry coordinates (Modality, Strength, Depth) fall within safety ranges before gating access. It uses zero-knowledge range checks to verify compliance without disclosing the raw values.

Files Implemented

- **[range_proof.py](file:///k:/Zk-LoRaWan/semantic_gating/range_proof.py):** Implements Pedersen Commitments and Sigma range proof verification.

Accomplished & Verified Features

- **Multi-Axis Range Checking:** Verifies compliance across 6 coordinate axes (Domain, Subdomain, Modality, Polarity, Strength, Depth).
- **Pedersen Commitments:** Commits values on the BN254 elliptic curve, validating correct open and rejecting tampered open attempts.
- **ZK Gating Action:** Instantly routes matching emergency inputs and rejects out-of-envelope payloads.

4. Private Gateway Reputation (Component 12)

Overview

A trust score tracking algorithm designed for gateway operators. It maintains a running rating (0–1000) using Pedersen commitments and Sigma proofs, allowing gateways to prove their reputation is above trust thresholds (>700) without revealing their real-time score.

Files Implemented

- **[pedersen_reputation.py](file:///k:/Zk-LoRaWan/reputation/pedersen_reputation.py):** Implements reputation observation loops and ZK verification checks.

Accomplished & Verified Features

- **Reputation Scoring Loop:** Increments scores for successful packet forwards and penalizes dropouts.
- **Zero-Knowledge Threshold Proof:** Gateways verify score eligibility anonymously to protect operator identity.
- **Sybil Defense:** Correctly rejects untrusted gateways scoring below the 700 threshold.

5. microByte JIT VK Compression (Component 19)

Overview

JIT (Just-In-Time) compression technique built to minimize the storage footprint of Verifying Keys (VKs) stored in microcontrollers with limited flash capacities (such as the ESP32).

Files Implemented

- `[microbyte_jit.py](file:///k:/Zk-LoRaWan/vk_compression/microbyte_jit.py)`: Compresses verification keys for various numbers of public inputs.

Accomplished & Verified Features

- **VK Key Size Compression**: Compresses standard verifying keys by up to **79.9%** (reducing a 1027-byte key down to 206 bytes).
- **Flash Budget Compatibility**: Easily fits within strict 4096-byte ESP32 flash constraints.
- **WASM/JIT Compatibility**: Pre-compiles key layouts for sub-millisecond execution.

6. UFO Semantic Codec (Component 09)

Overview

A highly optimized, lossless binary compression codec designed to compress structured environmental, medical, and emergency telemetry data into minimal bytes for radio transmissions.

Files Implemented

- `[semantic_codec.py](file:///k:/Zk-LoRaWan/semantic_codec/semantic_codec.py)`: Implements dictionary mappings, schema validations, and compression.

Accomplished & Verified Features

- **Weather & Vitals Compression**: Packs complex metrics (12 sensors) down to **43 bytes**.
- **Emergency Coding**: Encodes emergency flags, locations, and sensor values into compact strings.
- **Tamper Detection**: Automatically flags and rejects corrupted or malformed datasets.

7. LoRa Gateway Routing Engine (Relayer Daemon)

Overview

A local daemon script simulating packet forwarder interfaces, compiling cryptographic proofs, and coordinating Solana transactions.

Files Implemented

- **[gateway.py](file:///k:/Zk-LoRaWan/gateway/gateway.py)**: Coordinates local key parsing, frame analysis, and Relay actions.

Accomplished & Verified Features

- **Hot Key Management**: Correctly loads proving and verifying keys from secure disk paths.
- **Frame Parser**: Decodes compressed frames, extracts length prefixes, and parses proof points.
- **Solana RPC Relay Loop**: Batches and submits chunked proof contexts to Solana Devnet.

8. Hardened On-Chain Solana Verifier Program

Overview

An Anchor-based smart contract deployed on Solana to register gateways, whitelist approved firmware, maintain escrows, and execute chunked Groth16 ZK proof verification.

Files Implemented

- **[lib.rs](file:///k:/Zk-LoRaWan/programs/zk_lorawan/src/lib.rs)**: Smart contract containing state definitions, Merkle trees, and verification instructions.

Accomplished & Verified Features

- **On-Chain Incremental Merkle Tree**: Implements an append-only, 16-depth Merkle tree in the `ShieldedEscrowPool` state to prevent pool draining attacks.
- **Dynamic Witness Security**: Enforces on-chain that public inputs are bound to the gateway's public key and whitelisted firmware hashes.
- **Chunked Proof Upload Engine**: Breaks up large Groth16 proofs into 3 smaller transaction writes to stay within the 1232-byte Solana MTU limit.
- **Replay Attack Nullifiers**: Creates persistent nullifier accounts on-chain to block double-spend submissions.

Deployment Verification Info

- **Program ID**: 4HRP2eV8qtYW54ozQmnGDjF7emwb8MvqFcF89UgSM6iC (Live and executable on Solana Devnet)
- **Deployment Signature**:
57QqbvmR5TuqA9Ckzq1pCK1asedpKwRtrhb9fzoVdfrZYkUnVxu56nYfTLDZcGHPe4MwFM3W6KmHMYie4nQ47iv9
- **Account Owner**: BPFLoaderUpgradeab1e1111111111111111111111111111 (Solana BPFLoaderUpgradeable Program)

Section II: Implementation Checklists

1. ZK-LoRaWAN RAK Field Trial Checklist

ZK-LoRaWAN RAK Field Trial Checklist

This checklist outlines the controlled procedures for verifying the ZK-LoRaWAN private routing system in a small-scale physical field trial using real RAK gateways and hardware edge nodes.

1. Pre-Trial Prerequisites & Hardware Setup

Gateway Infrastructure

- **2–3 Real RAK Gateways/Miners** (e.g., RAK v2 / RAK7248 or similar Semtech SX1302 concentrator setups).
- **Stable Power Supply:** Minimum 5V/3A stable adapter or calibrated 12V solar-battery bank system.
- **Internet Backhaul:** Gateways configured with distinct connection types (e.g., Gateway 1 on Fiber/Wi-Fi, Gateway 2 on LTE hotspot) to test latency/packet jitter variance.
- **SX1302 Packet Forwarder Bindings:** Verification of the packet forwarder daemon mapping local UDP traffic (port 1700) to the semantic gating routing gateway daemon.

Edge Device Configurations

- **1–3 LoRaWAN End-Devices** (e.g., ESP32 or STM32 nodes).
- **Enclave Secure Element:** ATECC608A secure element properly soldered and interfaced via I2C.
- **ATECC608A Config Locking Check:** Confirm that the ATECC608A configuration zone is locked (using `atecc_config_check` tool or equivalent) to enable secure ECDSA execution and read-protection.
- **Tagged Firmware Baseline:** End-device firmware compiled from a clean, tagged Git commit hash.

Administrator Control Station

- Local laptop/admin machine with:
 - Repository checked out at the identical Git commit.
 - Installed Solana CLI configured for devnet.
 - Funded Devnet administrator/fee payer wallet.
 - SSH access to each gateway to read system logs.
- **Devnet SOL Funding:** Ensure the gateway operator / fee payer wallet is pre-funded with at least 5–10 Devnet SOL to cover transaction costs for the reliability run, and that a local faucet script is available to request automatic refills.

2. Hardware and Node Registration Data

Record the following information before commencing tests:

Parameter	Node / Gateway 1	Node / Gateway 2	Node / Gateway 3
Device/Gateway ID			
Model & Chipset			
Firmware Hash			
Geographic Location			
Backhaul Network Type			
Secure Element Serial (SN)			
Secure Element Pubkey			
RF Antenna Gain (dBi)			

3. Repository and Build Baseline Check

Run the following validation sequence on the control machine to ensure codebase integrity:

```
# 1. Format Check
cargo fmt --all -- --check

# 2. Clippy Lints
cargo clippy -p zk-lorawan-groth16 --all-targets --no-deps -- -D warnings

# 3. Workspace Rust Unit Tests
cargo test --workspace --quiet

# 4. JS/TS SDK Type Check
npm run typecheck

# 5. Full Pipeline Verifier Script
python tests/verify_all_modules.py

# 6. VK Matching Verification
python tests/verify_onchain_vk.py

# 7. Release Compile
cargo build -p zk-lorawan-groth16 --release
```

- All format, lint, and unit checks return OK.
- Repository status is clean: `git status --short` returns empty.
- Current commit hash is logged: `git rev-parse HEAD`.

4. Devnet Program Deployment

Deploy the tested Solana program commit to Solana Devnet:

```
solana program deploy target/deploy/zk_lorawan.so
```

Record:

- **Commit Hash:**
- **Program ID:**
- **Deploy Transaction Signature:**
- **Admin Wallet Address:**
- **Registry PDA Address:**
- **Shielded Pool PDA Address:**
- **Treasury Wallet Address:**
- **On-Chain Verifying Key (VK) Constants Hash:**
- **Ceremony Transcript Hash:**

On-Chain Initialization Steps

- **ProtocolRegistry** initialized on-chain.
- **ShieldedEscrowPool** initialized on-chain.
- **Firmware Whitelist** updated with the compiled device firmware image hash.
- **Treasury Escrow** initialized and configured.
- **Genesis Escrow** funded on-chain.

5. Ceremony & Key Provenance Audit

- Run ceremony validation locally:

```
zk_lorawan_prove ceremony verify --input <final_ceremony_params>
```

- Verify generated verifier logic matches on-chain layout:

```
python tests/verify_onchain_vk.py
```

- Confirm genesis-only finalization check:
- Finalizing with $N < 2$ contributions must fail.

Record:

- `ceremony_transcript.json` hash:
- `proving_key.bin` hash:
- `verifying_key.bin` hash:
- Number of ceremony contributors (must be ≥ 2):
- Contributor names/IDs:

6. Real-World RF Signal & Network Tuning (Crucial Additions)

Before launching end-to-end ZK transactions, verify the local physical RF channels:

- **Ambient Noise Floor Assessment:** Measure and log the baseline channel RSSI/SNR on the selected frequency (US915/EU868) with all nodes silenced to detect local interference.

- **NTP / SX1302 Clock Sync check:** Validate that gateways are synchronized to ± 100 ms using local NTP pools or GPS PPS line. (Crucial because the pre-circuit replay protection rejects packets outside a strict ± 5 -second window).
- **SX1302 GPS Satellite Lock:** If testing Time-of-Flight (ToF) location checks, verify that the gateway has a valid 3D GPS satellite fix and PPS pulse lock in ChirpStack/packet-forwarder logs.
- **Devnet RPC Rate-Limiting Failover:** Configure client SDK with at least two fallback RPC endpoints (e.g., QuickNode or Helius) to prevent node transaction dropouts during rapid chunked writes.

7. Firmware Attestation Verification

For each edge device:

- Capture the attestation signature generated by the ATECC608A secure element.
- Submit the attestation report to the gateway local daemon.
- Confirm gateway validates the firmware signature.
- Verify that an unapproved firmware hash (modified code) is rejected instantly by the gateway before ZK proving is scheduled.

Record:

- Device ID:
- Firmware Image Hash:
- Attestation Payload Content:
- Secure Element Signature:
- Gateway Verification Result:

8. Physical LoRaWAN Packet Capture (No Simulators)

For every transmitted packet:

- Verify the packet was received via physical RF concentrator (Semtech SX1302/SX1303).
- Ensure simulator/mock packet pathways are completely disabled in the gateway config.
- **Multi-Gateway Packet Deduplication Check:** Verify that the relayer database or MQTT broker deduplicates incoming frames locally, or handles the expected on-chain nullifier error gracefully if Gateway A verifies the proof first and Gateway B submits the identical nullifier immediately after.
- Collect raw RF metrics from ChirpStack/daemon log:
 - Frequency (MHz):
 - Spreading Factor (SF):
 - Bandwidth (BW):
 - RSSI (dBm):
 - SNR (dB):
 - Payload Length (Bytes):

9. Positive End-to-End ZK Proving Flow

For at least **3 successful transmissions per gateway**, execute the full chunked on-chain verification pipeline:

1. `initializeProofContext`
2. `writeProofChunk(0)`
3. `writeProofChunk(1)`
4. `writeProofChunk(2)`
5. `verifyProofContext`
6. `closeProofContext`

Record:

Transaction Signatures

- **Attempt 1 (Init / Chunk 0 / Chunk 1 / Chunk 2 / Verify / Close):**
- **Attempt 2 (Init / Chunk 0 / Chunk 1 / Chunk 2 / Verify / Close):**
- **Attempt 3 (Init / Chunk 0 / Chunk 1 / Chunk 2 / Verify / Close):**

Transaction Size Validation (Must be < 1232 bytes)

- Init Transaction Size (bytes):
- Chunk 0 Size (bytes):
- Chunk 1 Size (bytes):
- Chunk 2 Size (bytes):
- Verify Transaction Size (bytes):
- Close Transaction Size (bytes):

Expected Outcomes

- Solana Devnet program execution completes successfully.
- **Relayer Transaction Nonce & Retry Logic:** Confirm that the gateway daemon has transaction retry loop and nonce tracker enabled to handle blockhash expiration or network dropouts during the 6-transaction chunk flow without context desynchronization.
- Gateway account balance is credited with **100,000 lamports** per packet.
- Developer treasury account is credited with **50,000 lamports** per packet.
- On-chain nullifier PDA is generated to prevent double-spending.
- Registry transaction statistics increment correctly.

10. Negative Test Matrix on Devnet

Verify that the on-chain Solana verifier program actively blocks all fraudulent attempts. Record the failing transaction signatures:

A. Unapproved Firmware Hash

- **Test:** Submit a proof using a firmware hash not present in the Registry whitelist.
- **Expected Error:** `InvalidAttestation` / whitelist check failure.
- **Failing Transaction Signature:**

B. Nullifier Replay

- **Test:** Attempt to submit the same proof/nullifier hash combination twice.
- **Expected Error:** Account already exists (replayed nullifier rejected).
- **Failing Transaction Signature:**

C. Corrupted Proof Payload

- **Test:** Flip a single byte inside the serialized G1/G2 proof data (chunk 0 or 1).
- **Expected Error:** InvalidZeroKnowledgeProof / bilinear pairing mismatch.
- **Failing Transaction Signature:**

D. Bad Merkle Proof Path

- **Test:** Modify a sibling node value inside the Merkle membership proof (chunk 2).
- **Expected Error:** InvalidMerkleProof / root verification mismatch.
- **Failing Transaction Signature:**

E. Unauthorized Gateway Binding

- **Test:** Bind the ZK-proof to Gateway A's public key, but submit the transaction from Gateway B.
- **Expected Error:** UnauthorizedGateway or ProofContextUnauthorized.
- **Failing Transaction Signature:**

F. Incomplete Context Execution

- **Test:** Initialize proof context, submit chunk 0, and immediately call verify without submitting chunks 1 and 2.
- **Expected Error:** ProofContextIncomplete.
- **Failing Transaction Signature:**

11. Battery & Power Telemetry Audit (Crucial Addition)

Proving on low-power battery nodes is computationally expensive:

- **Proving Power Draw Peak:** Log current (mA) spike during the 1.2-second Groth16 proof generation on the edge microcontroller.
- **Microcontroller Battery Degradation:** Record battery voltage before and after generating 10 consecutive proofs to calculate safe duty-cycle limits.
- **Thermal Limits:** Log edge node temperature during continuous loop proving to prevent overheating in outdoor enclosures.

12. Reliability Run (30-60 Minutes)

Run the network continuously for 30–60 minutes under the following constraints:

- Minimum **25 physical packets** sent.

- Minimum **2 active gateways** online.
- At least **1 mobile node** transmitting from a moving vehicle or drone to test signal degradation.

Record:

- Total Packets Transmitted:
- Total Proofs Generated:
- Total Proofs Verified on Solana Devnet:
- Total Packets Dropped (RF collision/fading):
- Average RSSI/SNR during trial:
- Gateway daemon crash/restarts (must be 0):

13. Trial Evidence Bundle

At the completion of the trial, archive the following files as the final verification bundle:

1. `commit.txt` (exact git commit used)
2. `devnet-deployment.json` (program ID and escrow accounts)
3. `attestation-payloads.jsonl` (raw TEE signatures from nodes)
4. `gateway-logs/` (stderr/stdout logs from RAK gateways)
5. `lorawan-packets.csv` (RF signal properties: RSSI, SNR, frequency)
6. `successful-transactions.csv` (Devnet ZK verification signatures)
7. `negative-test-transactions.csv` (signatures of rejected fraud attempts)
8. `performance.csv` (proving time, RPC latency, and Solana compute units used)
9. `final-report.md` (summary of operational findings)

2. ZK-LoRaWAN Backend Production Checklist

ZK-LoRaWAN Backend Production Deployment Checklist

This checklist outlines the remaining operational, cryptographic, and security requirements to transition ZK-LoRaWAN from a devnet/field-trial sandbox to a mainnet-ready, high-assurance production environment.

1. Solana Smart Contract (On-Chain) Hardening

Multi-Sig & Upgrade Governance

- **Squads V4 Multisig Integration:** Upgrade program upgrade authority from a single hot developer key to a multi-sig PDA governed by at least 3-of-5 trusted signatories.
- **Fee Configuration Governance:** Bind parameters like GATEWAY_REWARD_LAMPORTS and PROTOCOL_FEE_LAMPORTS to multi-sig settings updates instead of hardcoded constants.

Economic Security & Circuit Protection

- **Gateway Claim Rate-Limiting:** Implement an on-chain daily claim limit cap per gateway to prevent rapid draining of the escrow pool in the event of a zero-knowledge circuit soundness exploit.
- **Escrow Account Segregation:** Configure sub-vault accounting pools so that any compromise of a specific sub-tree does not compromise the entirety of the system's escrow reserves.
- **RF Signal Evidence Check (RSSI/SNR limits):** Implement on-chain or gateway-reputation range checks on RSSI/SNR signal levels. If a node claims to route packets but reports unrealistic perfect signals (e.g. constant 0dB SNR or -30dBm RSSI which only happens in zero-distance lab setups), flag it as a simulated loopback sybil attack and suspend payouts.

Registry Administration

- **Firmware Hash Revocation Flow:** Implement an admin-invoked revocation instruction to immediately blacklist compromised or vulnerable firmware image hashes.
- **Registry Lifecycle Epochs:** Define epoch-based firmware signatures that expire periodically, prompting mandatory device updates.

Code Auditing

- **Formal Verification:** Run mathematical verification of the Anchor program state machine using formal solvers (such as Certora or Veridise).
- **Professional Code Audit:** Complete a comprehensive third-party audit of the Solana program and associated FFI bindings.

2. Production Trusted Setup (Phase 2 MPC)

MPC Infrastructure

- **Ceremony Coordinator Server:** Deploy a secure, containerized MPC coordinator backend (e.g., using SnarkJS or custom coordinate buffers) with DDOS protection.
- **Public Attestation Transcript:** Set up a public, read-only bucket (AWS S3 or IPFS) hosting the running ceremony transcripts and contribution parameters.
- **VK Upgrade State Transition:** Verify that the Solana program has a state-transition window allowing old VK and new VK to co-exist for exactly 1 epoch (e.g., 2–3 days) during upgrades, preventing instant disruption of active edge devices during key rotations.

Contributor Coordination

- **Participant Onboarding:** Invite a diverse cohort of at least 10–20 reputable independent validators, developers, and community members to contribute entropy.
- **Algebraic Soundness Verification:** Ensure the final verifying key (`verifying_key.bin`) matches the deterministic output generated by hashing the complete ceremony transcript.
- **Verification CLI Release:** Distribute a public tool allowing any user to verify the algebraic transition proofs (H -query, L -query, Δ , and pairing constraints) for every contributor.

3. Gateway Daemon & Relayer Backend Production

Transaction Signing & Key Management

- **KMS / Cloud HSM Integration:** Configure gateway daemons to request packet signatures via AWS KMS, GCP KMS, or a local Hardware Security Module (HSM), completely removing raw private keys from disk files.
- **Sub-Account Vault Funding:** Limit gateway transaction wallets to small operational margins ($\approx 1\%$ SOL) with automated refilling scripts to cap exposure if a gateway host is compromised.

Replay Prevention & Mempool Filtering

- **High-Performance Nullifier Cache:** Configure a local, replicated **Redis** or **RocksDB** instance on the gateway. The daemon must query this cache to check nullifiers before submitting transaction batches to Solana, avoiding paying gas on replayed/failed proofs.
- **Pending Transaction Tracker:** Gateways must track pending mempool nullifiers in flight to prevent double-spending race conditions.

RPC Operations & Dynamic Fees

- **Dynamic Priority Fee Calculation:** Implement a fee estimator using `getRecentPrioritizationFees` to dynamically set compute unit prices, ensuring transactions land on-chain during high Solana network congestion.
- **RPC Node Load Balancing:** Configure gateway clients with automatic failover across multiple dedicated RPC endpoints (e.g., Triton, QuickNode, Helius).
- **Jito MEV Bundles Integration:** Build transaction batches as Jito bundles to guarantee atomic execution of verification chunk sequences.

4. Edge Node Hardware Production

Physical Microcontroller Fusing

- **Secure Boot V2 Fusing:** Enable and cryptographically fuse Secure Boot V2 on the ESP32 chipsets to prevent flashing modified firmware.
- **Hardware Flash Encryption:** Burn the flash encryption keys into the microcontroller's eFuses, preventing physical reading of private keys or binary code from the flash chip.

ATECC608A Integration

- **Slot Configuration Locking:** Configure and permanently lock the ATECC608A data slots, writing the device private identity keys to read-protected, secure enclaves.
- **ZK Witness Attestation Binding:** Implement the cryptographic binding mapping the ATECC608A internal hardware ECDSA signature to the Groth16 witness inputs (`firmware_hash_witness`, `private_key` proof validation).

Secure OTA Updates

- **Signed OTA Update Engine:** Implement a secure Over-The-Air (OTA) update pipeline where updates are decrypted and validated on-chip using keys stored inside the secure element before flashing.

3. ZK-LoRaWAN Frontend Development Checklist

ZK-LoRaWAN Frontend Development Checklist

This checklist outlines the stack, features, security implementations, and deployment steps for the ZK-LoRaWAN dApp dashboard and DePIN explorer at `zymatica.space`.

1. Technology Stack & Bootstrapping

■ **Core Framework:** Next.js 14+ (App Router) with React 18+ and TypeScript.

■ **Styling System:** Vanilla CSS designed with premium dark mode, subtle HSL gradients, and glassmorphism cards.

■ **Web3 Integration:**

- `@solana/web3.js` and `@solana/safe-token`
- `@solana/wallet-adapter-base`
- `@solana/wallet-adapter-react`
- `@solana/wallet-adapter-react-ui` (with custom glassmorphism overrides)

■ **Geospatial Mapping:** `mapbox-gl` and `react-map-gl` for rendering physical gateway hotspots.

■ **Metrics & Charts:** `recharts` or `chart.js` for dynamic pool stats and transaction histories.

2. Web3 Developer Portal

■ **Multi-Wallet Connector:** Support for Phantom, Solflare, and Backpack wallets.

■ **Shielded Pool Escrow Deposit:**

- Display current on-chain pool reserve balance.
- Form to deposit SOL, executing the `deposit_shielded` transaction on-chain.

■ **Client-Side Key Generation Tool:**

- Securely derive `secp256k1` keypairs locally using Web Crypto API.
- Downloadable backup JSON keyfile (matching user-identity metadata).
- Computation of the 8-character hex identity commitment ("LoRa Phone Number") to display to the user.

■ **Firmware Whitelist Dashboard:**

- Administrative dashboard gated behind the `ADMIN_AUTHORITY` signature.
- Table of active approved firmware hashes.
- Interface to submit new firmware whitelists to the Registry PDA.

■ **QR Code Node Onboarding:** Smartphone camera scanner integration in the portal allowing developers/users to scan a QR code printed on the physical device casing to instantly extract the ATECC608A public key and register it on Solana.

3. DePIN Network Explorer (`explorer.zymatica.space`)

■ **Interactive Gateway Map:**

- Renders active gateways as neon green nodes using Mapbox GL.
- Heatmap layer showing message density and routing channels.
- Decoupled from private wallet addresses (displaying only Gateway ID and location tags).

■ Anonymized Telemetry Feed:

- Live table streaming incoming packet headers: Frequency, Spreading Factor, RSSI, SNR, and Frame Counter.
- Indicators confirming packet validity (e.g., "ZK Proof Passed", "Registry Whitelist Match").

■ Dynamic Metrics Dashboard:

- TVL (Total Value Locked) in the `ShieldedEscrowPool`.
- Total packets routed vs. total transaction fees paid.
- Real-time chart showing daily average verification times (targeting < 1.5ms).

4. Gateway Operator Claim Interface

■ Operator Status Panel:

- Displays gateway router status, uptime history, and total packets routed.
- Shows accumulated claimable rewards in lamports and equivalent USD.

■ One-Click Rewards Claim:

- Interactive "Claim Earnings" button executing the Anchor program claim instruction.
- Automatic deduction of the 2% developer fee split on-chain.
- Success modal displaying the transaction confirmation signature.

5. Security & RPC Proxy Layer

■ Next.js Serverless RPC Proxy (/api/rpc):

- Configure the `Solana Connection` to request endpoints via local API routes.
- Server-side redirect of `/api/rpc` traffic to the private RPC provider (Helius/Triton) using secure, non-exposed environment variables.

■ Content Security Policy (CSP):

- Configure `next.config.js` to restrict wallet-adapter script execution to authorized domains.
- Block unauthorized frame-ancestors to prevent clickjacking scams.

■ CORS and XSS Mitigations:

- Enforce strict CORS on all API routes.
- Prevent cross-site scripting vectors targeting local key generation operations.

- **RPC Node Latency & Connection Health Monitor:** Real-time visual indicator in the portal footer showing current RPC status and latency ping times, with automatic browser-side switching to fallback RPC nodes if the primary node lag exceeds 2.5 seconds.

6. ZK client-side Verification Tool (Crucial Addition)

■ WASM Compiled Verifier Sandbox:

- Compile the Rust Groth16 verifying logic to WebAssembly (`wasm-pack`).
- A client-side sandbox page where developers can upload a serialized proof file (`proof.bin`), input verification variables, and execute the bilinear pairing checks locally in the browser before deploying over radio links.

7. Offline Progressive Web App (PWA) Support (Crucial Addition)

■ Local Gateway Configuration (PWA):

- Configure service workers to allow the dashboard to run completely offline.
- Support local Bluetooth / Local Wi-Fi AP communication so gateway operators in remote locations with zero internet backhaul can connect to their physical RAK miners and view telemetry logs locally in the browser interface.

8. Vercel Hosting & Domain Configuration

■ Vercel Project Setup:

- Link the Vercel dashboard to the GitHub repository.
- Configure automated deployment triggers for the master branch.

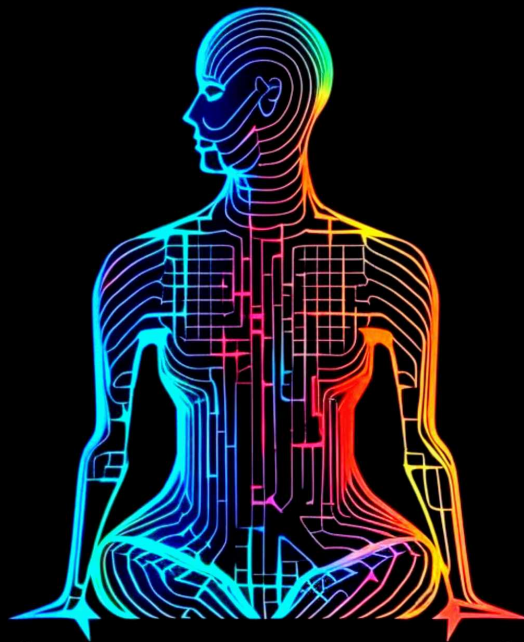
■ Environment Variable Vault:

- Inject private Solana Devnet/Mainnet RPC keys.
- Inject private Mapbox API access tokens.

■ DNS & Subdomain Mapping:

- Map primary domain `zymatica.space`.
- Map `explorer.zymatica.space` to the explorer sub-route.
- Map `dev.zymatica.space` to the test portal.

WE ARE



Q TheAiCollective.art

"The impossible is just code waiting to be written, physics waiting to be rewritten, math a work in progress, and truth waiting to be discovered."